Alabama PM2.5 prediction

Pooled county study | Findings and methods | 11 October 2026

Scope: the completed pooled study using ten county worksheets and six forecasting models. This report summarizes the saved experiment; it does not introduce new fitting or analyses.

Main finding

LSTM was strongest in chronological validation. CNN-LSTM had the lowest October-December test error, but its small advantage over LSTM was uncertain. The experiment does not establish that adding a convolution layer reliably improves prediction.

Measure	Result
Merged dataset	3,650 county-days; 10 counties; daily observations in 2025
Clean PM2.5 observations	3,580 observed; 70 missing after cleaning
Shared test observations	887 county-days, October-December
CNN-LSTM test MAE	2.113 micrograms per cubic meter
Improvement over persistence	Approximately 14.7% lower test MAE
Hybrid versus LSTM	MAE difference -0.0201; 95% interval [-0.0558, 0.0088]

Dataset and cleaning

The county sheets were combined into a panel retaining date, county, measurement values, source sheet and source row. Raw observations remain available. Negative PM2.5 readings (11) were set to missing in the analysis copy, alongside 59 originally missing readings; targets were never imputed.

County labels and dates were standardized and duplicate keys checked. Joint-zero temperature/dew point/humidity placeholders and invalid weather values were marked missing. Positive extremes were retained. Feature medians, scaling and missingness indicators were fitted using each training period only, with no future interpolation.

What pooling means

Models learn from all counties while each 14-day history stays within its own county. County identity is an input. Pooling supplies more rows, but still only one year and one annual seasonal cycle. Monitor measurements are site observations within counties, not county-wide spatial averages.

How the comparison was validated

The target is the next day's PM2.5. Inputs include past PM2.5 and weather, missingness, county identity and calendar features. No same-day or future weather is used. All learned models predict a correction to previous-day persistence.

Fold	Outer training	Outer validation
1	January-March	April-May
2	January-May	June-July
3	January-July	August-September

These are three expanding-window time-series folds, rather than shuffled k-fold splits. All counties share calendar boundaries. Earlier observed validation days may enter later histories: this evaluates daily one-step forecasts, not a multi-month recursive forecast.

Tuning and early stopping

Each model family is selected by its lowest unweighted mean validation MAE across the three folds. Neural early stopping uses the final 21 days inside the outer training period, with separate inner preprocessing. The model is then reinitialized and refitted on the full outer training segment for the selected duration before scoring validation.

Model	Search and selected settings
Persistence	Previous-day concentration; no parameters to tune
Ridge	Alpha: 1, 10, 100, 1000; selected 1000
Random forest	Depth 6/unlimited and leaf size 5/15; selected depth 6, leaf 15; 150 trees
CNN, LSTM, CNN-LSTM	Width 8/16 and learning rate 0.001/0.0003; each selected width 8, rate 0.0003; dropout 0.1

Stochastic families average seeds 11, 22 and 33, including random forest. Final neural epoch counts are each seed's median inner-stopping duration across folds. Maximum inner training is 60 epochs, patience 8, batch size 64. All settings were frozen before final January-September fitting and October-December scoring.

Fairness and verification

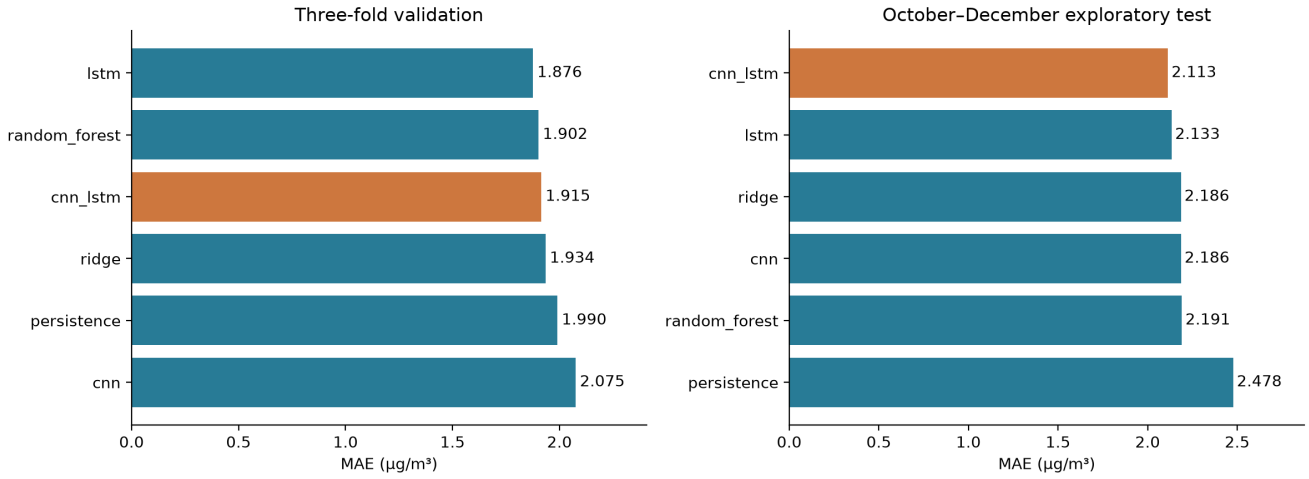
All six models use the same eligible test county-days. Cases with missing targets or missing previous-day PM2.5 are excluded; older missing inputs receive training-only imputation. The selected configurations, per-seed results, training histories, fitted models and data hashes are saved.

Seven pipeline tests and 11 independent audit checks passed, including chronological windows, training-only preprocessing, model selection, recomputed metrics and seed averaging. Checkpoint reload predictions were verified. CPU execution was tested on native ARM64; Apple MPS support is included but was unavailable in this session.

Overall model results

Lower MAE and RMSE indicate smaller errors. Values below use micrograms per cubic meter. CV MAE is the mean across selected validation folds (1,783 validation cases total); test results use 887 shared county-days.

Model	CV MAE	Test MAE	Test RMSE	Test R-squared
CNN-LSTM	1.915	2.113	2.734	0.355
LSTM	1.876	2.133	2.750	0.347
Ridge	1.934	2.186	2.823	0.312
CNN	2.075	2.186	2.870	0.289
Random forest	1.902	2.191	2.808	0.319
Persistence	1.990	2.478	3.299	0.060



Does convolution add value?

CNN-LSTM minus LSTM test MAE is -0.0201, with a paired 95% block-bootstrap interval of [-0.0558, 0.0088]. Because it includes zero, their average test errors are not clearly distinguished. The hybrid's apparent advantage is about 0.9% relative to LSTM. LSTM remains a simpler competitive starting point.

The interval uses 1,000 circular seven-day calendar blocks, keeping counties together. It conditions on fitted ensembles and is pointwise, without multiple-comparison adjustment. Architecture-specific loss functions, capacities and search budgets also differ, so this is not a controlled isolation of architecture alone.

Average accuracy does not imply peak accuracy

High days exceed each county's January-September 95th percentile. Only 27 eligible test cases meet this definition. Persistence had the lowest high-day MAE (3.674), compared with CNN (3.872), ridge (4.312), random forest (4.370), CNN-LSTM (4.435) and LSTM (4.448). A model selected for average error may not be best for pollution alerts.

County findings and conclusions

County	Test n	Lowest MAE model	MAE	CNN-LSTM MAE
Baldwin	92	CNN-LSTM	1.833	1.833
Clay	88	LSTM	1.750	1.776
DeKalb	84	CNN-LSTM	2.229	2.229
Etowah	88	Random forest	2.777	2.914
Jefferson	92	LSTM	3.243	3.258
Mobile	88	Ridge	1.816	1.837
Montgomery	89	CNN-LSTM	1.933	1.933
Morgan	86	CNN-LSTM	1.935	1.935
Sumter	92	LSTM	1.613	1.631
Tuscaloosa	88	CNN-LSTM	1.770	1.770

CNN-LSTM had the lowest observed county MAE in five counties, LSTM in three, random forest in Etowah and ridge in Mobile. Jefferson had the largest errors, while Sumter had the smallest. These are descriptive county rankings, not separate significance tests.

For the coastal counties, CNN-LSTM MAE was 1.833 in Baldwin and 1.837 in Mobile. Baldwin favored the hybrid; Mobile favored ridge (1.816). Both coastal counties performed better than the pooled hybrid MAE, but location alone is not established as the cause. Baldwin and Mobile share weather inputs, limiting their independence.

What can be concluded

1. Pooling counties supports a useful common prediction model without losing county-specific histories. All learned families improved overall test MAE over persistence.
2. LSTM is a defensible starting point based on validation; a reliable extra benefit from CNN-LSTM has not been demonstrated.
3. Model performance varies by county and objective. Pollution peaks require a separate, explicitly designed evaluation.

Limits and next research steps

October-December outcomes were inspected in earlier studies, so this test is exploratory. Selected CV scores participated in tuning and are not unbiased nested-CV estimates. One year limits seasonal generalization; nearby counties and shared weather create dependence. Weather provenance, pressure/rain definitions and real-time availability remain partly unresolved.

The strongest next evidence would come from untouched additional years. Further experiments could compare pooled versus county-specific models on identical splits, add past neighboring-county PM2.5, and optimize peak-event prediction with a prespecified metric. These experiments have not been performed here.

Evidence: pooled_county_study/results/{pooled_test_metrics.csv, cv_selected_summary.csv, county_test_metrics.csv, paired_mae_intervals.csv, selected_settings.json, independent_verification.json}; data/{cleaning_log.csv, workbook_verification.json}. Full code and results are in the saved study folder and downloadable ZIP.